
Story Sprout: A Picture-Book Studio Where Children Direct the AI and Keep the Pen

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Abstract

1 Story Sprout is a browser-based picture-book studio for children ages 7–9, built
2 around an explicit allocation of agency to the child users, which is enforced in the
3 system. The children originate every sentence of the story and every illustrated
4 scene detail they specify. The children request each illustration and decide alone
5 what will be in the book. The model’s creative-production role is confined to draw-
6 ing the illustration and redrawing it based on the child’s requests for changes. The
7 system design began in a community-center program, where facilitators watched
8 children’s descriptions grow more specific as LLM-generated pictures never quite
9 matched the child’s imagined scene. The studio makes that mismatch the center
10 of the activity: the children describe, compare with the scene they imagined, then
11 accept or revise by saying more, more precisely. The system asks the child whether
12 the picture matches what they imagined. It never makes that judgment for the child.
13 The finished work and its revision record live in a local file held by the supervising
14 adult: no child accounts, no readable server-side project copy, an encrypted class
15 relay the platform cannot read. The goal of the activity is to allow children to learn
16 through making real things for a real audience, following Papert’s constructionist
17 tradition. In what follows, we present the agency framework, the deployed system,
18 its layered safety mechanism, the aspects of the agency allocation that could be
19 contested, how our system compares to what is currently available, and how our
20 workshops enabled us to engage in iterative child-centered design.

21 1 Introduction

22 Consumer generative-AI tools tend to collapse creative work into a transaction: you request, it
23 creates. Research on collaborative learning suggests that cognitive growth can arise from interaction
24 processes such as explaining, negotiating, and refining ideas, rather than from simply receiving an
25 output [Dillenbourg et al., 1996; Dillenbourg, 1999]. A recent review finds that agency in child-
26 AI interaction is rarely defined explicitly [Voysey et al., 2026]. When the maker is a child, that
27 arrangement is worth questioning: a nine-year-old who can summon a polished illustration in four
28 seconds has been handed enormous production capacity. Papert placed learning in the gap between
29 what a maker intended and what they got [Papert, 1980]. Yet near-instant results leave little room for
30 debugging. Our goal was not to make the generation better, but to help children’s creativity grow
31 through interaction with the generative AI. That raises the question this paper addresses: which parts
32 of making a picture book should the model take on, and which should remain with the child?

33 Story Sprout¹ answers this question with an explicit allocation of agency, set by the system’s
34 constructionist goals (§2) and enforced in code. The model is an illustrator under contract: asked to
35 render from the child’s description, never to originate the story, decide what should be illustrated, or
36 judge the result. Describing is the child’s job, and so is judging whether the picture is good. Story

¹Code: https://github.com/kmicodbase/kmi_story_sprout_studio. The studio runs at https://kmicodbase.github.io/kmi_story_sprout_studio/workshop-plugin.html.

37 Sprout was built for, and piloted in, an after-school program at a community center: cohorts of thirty
38 to forty children aged 7–13 (target band 7–9, §5), borrowed devices, no child accounts.

39 **Contributions.** (1) An explicit agency allocation for child-facing creative-AI systems. Table 1
40 formalizes the allocation and serves as an audit template for other child–AI creative systems. (2)
41 Story Sprout, a deployed studio that implements the allocation through a book editor, Pip the
42 illustrator, layered safeguards, a local file, and a paired record of the child’s words and the composed
43 prompt (§3).

44 **Related work.** Wordcraft lets adult writers work with a model to continue, fill in, or rewrite their
45 text [Yuan et al., 2022]. Mathemyths takes turns telling the story with a child [C. Zhang et al.,
46 2024]. StoryDrawer turns a child’s spoken story into drawings and can suggest related sketches
47 [C. Zhang et al., 2022]. StoryBuddy distributes interactive story reading and question-answering
48 across parent, child, and AI [Z. Zhang et al., 2022]. Narratron augments hand-shadow puppetry with
49 AI-generated storytelling and co-performance [Zhao & Bao, 2023]. Colin asks children to contribute
50 to an AI-expanded story through speech and drawing, with facilitator oversight [Ye et al., 2025].
51 The closest system to the current work is StoryPrompt: children write the story while the AI offers
52 keywords and illustrations [Fan et al., 2025]. Story Sprout builds on prior work but differs in keeping
53 story authorship and scene-content decisions with the child, by limiting the model to rendering those
54 decisions (§2).

55 2 Design rationale and the allocation of agency

56 Constructionism proposes that children learn through making personally meaningful artifacts that
57 can be shared with others [Papert, 1980; Papert & Harel, 1991]. The curriculum and, later, the
58 studio apply this principle through picture-book creation, in which children make decisions about a
59 complete artifact intended for a real audience. Papert also emphasized debugging as part of learning:
60 an unexpected result can provide information about what needs to be reconsidered or revised rather
61 than marking failure [Papert, 1980]. Story Sprout treats a mismatch between the intended scene
62 and the generated picture as a form of debugging. The mismatch may come from an omitted or
63 ambiguous description, or from the model itself, but the child’s available means of responding is
64 to clarify, restate, or revise the description. Delegating the rendering serves the same goal: rapid
65 visual production makes the consequences of language inspectable in tens of seconds, and gives a
66 seven-year-old an introduction to how generative AI functions. Together, these ideas provide the
67 learning-science basis for Story Sprout’s design.

68 In summer 2025, we ran the curriculum as an eight-session program: forty children collaborated on
69 original stories by hand, described each scene to public image-generation sites, repeatedly revised
70 their descriptions and requested new renderings, and chose which pictures would be printed in their
71 books. The result was five complete picture books. The finished books were physical artifacts that
72 children could read aloud and share with classmates and family members. The sessions yielded an
73 important observation: children’s scene descriptions became more specific, not through teaching
74 them how to prompt, but because the picture differed from the scene in the child’s head; whether
75 a detail had gone unsaid or the model was simply wrong, this led the child to say more, and more
76 precisely. Story Sprout Studio builds that pattern into the core loop (§3.1).

77 The studio formalizes the curriculum arrangement as an explicit allocation of agency, addressing the
78 gap identified in §1. Story Sprout separates two elements that discussions of agency often conflate:
79 decision authority and causal influence. In Story Sprout, authority over story text, requested scene
80 content, and image judgment and acceptance belongs to the child; image rendering is delegated to the
81 model; the supervising adult controls the project file; and the platform and provider retain authority to
82 block prohibited content. These allocations describe decision authority, not every source of influence
83 on the resulting work. Causal influence is broader: the renderer’s priors shape the pictures, the adult
84 facilitator frames the task, and a previously accepted picture can shape what a child writes next.

85 In practice, the allocation is enforced through four commitments, implemented in the code. 1. Every
86 word in the book is the child’s: the model neither generates, completes, nor improves story text,
87 not even for a stuck writer or a missing title (§3.2). 2. Rendering is the model’s role: the studio
88 offers no drawing tools; to get a picture, the child says what it should be (§3.2). 3. Judgment is the
89 child’s prerogative: after every generation, the system asks one plain question, "Is this the picture
90 you imagined?" It does not evaluate the child’s answer (§3.1). 4. The book is a single local file; it is

Table 1: The agency allocation.

Creative function or decision	Allocated authority or role	Other causal influences	Constraint
Story text	Child	An accepted picture can shape what the child writes next (§6)	No automated software pathway can write or edit story text
Requested scene content	Child	The scribe’s composition can deviate from the child’s input (§4.1)	The child’s words and the composed prompt are logged as separate strings (§3.3, §4)
Image rendering	Model, the sole delegated creative-production role	The platform’s content line, the renderer’s built-in guidelines (separate from our design), visual continuity from previous images, and the model’s own priors (§3.3, §6)	Preview before the page; every attempt kept side by side (§3.2)
Judgment, acceptance of image	Child	When used in a group, peers and a facilitator may help shape the illustrated story and its intended audience (§5)	Explicit accept step before a picture enters the book; removal only on request
Data custody	Supervising adult	Requests transit the provider under its API terms (§3.3)	A local project file; client-side encryption on the relay; no child accounts

91 shared only if the adult chooses (§3.3). Table 1 places these commitments alongside the other causal
 92 influences and the constraints that support each allocation.

93 Table 1 can be read as a template: for each creative decision, it identifies who holds authority, what
 94 other actors influence the outcome, and what mechanism supports the claim. These mechanisms
 95 provide different levels of assurance. Some are structural. For the story text, no code path allows AI
 96 to write or edit it. Others depend on instructions to a probabilistic model. Pip is instructed to preserve
 97 the child’s wording when a request for an image is received (example in §3.2), and it is turned into a
 98 prompt for the image generator but the generator may not always understand the child’s intention,
 99 allowing the child to examine the resulting image, and then re-describe to Pip what is being requested.
 100 The source of a mismatch between the child’s imagined scene and the rendered image matters because
 101 it determines whether the child should clarify the description, request another rendering, or recognize
 102 a limitation of the model. A generative renderer brings habits and random variation of its own, so not
 103 every mismatch traces to the child’s words. Two characteristics of generative image models are likely
 104 to persist even as their rendering capabilities improve: 1) the renderer will fill in whatever the image
 105 needs that the child’s description leaves out; 2) the renderer will base its interpretation of the child’s
 106 description on its own priors.

107 Story Sprout’s layered safety and local, account-free data design were motivated by two observations
 108 of existing commercial systems: public image-generation sites were designed for adults and lacked
 109 child-specific safeguards, while online storage of children’s work raised consent and governance
 110 concerns (§3.3).

111 3 The system

112 The platform is made up of several static HTML pages and a small proxy server. Everything the
 113 system retains about a child’s project lives in a file the supervising adult controls. Requests go to the
 114 model provider (OpenAI in this case) for generation, and the Story Sprout platform does not keep
 115 any readable copy of that data.

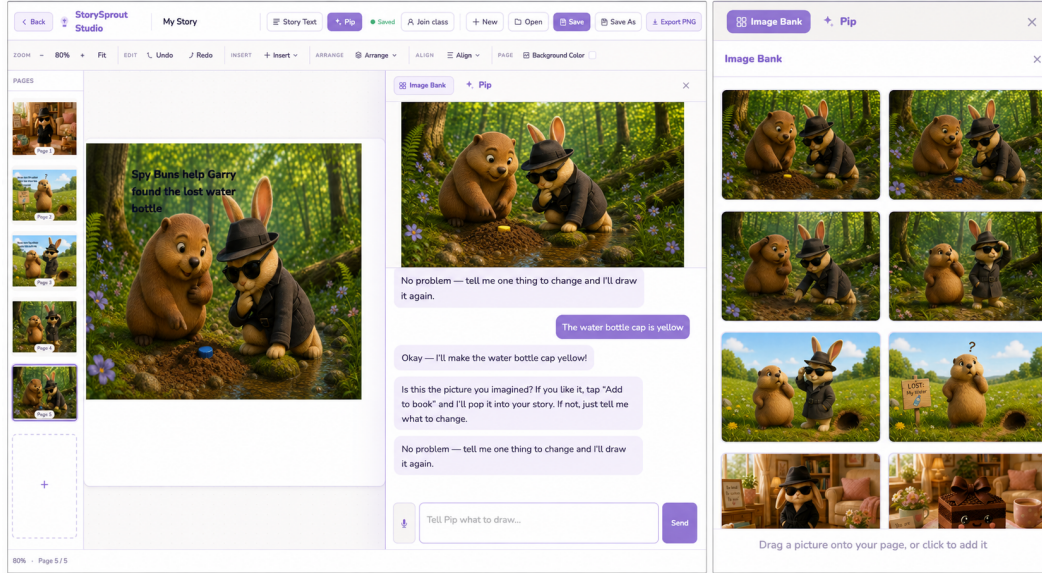


Figure 1: Deployed Story Sprout Studio interface. Left: Pip in action. Right: Image Bank.

3.1 The studio

The front end is a canvas-based book editor (pages, text boxes, shapes, layout controls, undo/redo, per-page PNG export) that stores projects locally (§3.3). For each illustration, the child describes the scene to be illustrated to Pip, the child’s virtual helper; once Pip receives enough detail about the image the child wants, Pip requests an image under its scribe contract (§3.2). The resulting image appears in a preview for the compare step. When the picture is not what the child wants, the child can tell Pip what is missing or what should be added, and the new attempt appears beside the old in the Image Bank (Figure 1) for easy comparison of the two. Generated images remain in the preview and Image Bank until the child selects the "Add to book" button.

For each image generation request, the platform only adds the one-line safety constraint (example in §3.2). A child can ask for watercolor or a comic book style, among others, and the theme persists throughout the rest of the book (§3.2). If the child does not name a style, the platform leaves that choice to the image model, which draws from the child’s description and its priors.

Before working unaided, the child is walked through the process. A picture can be described **by text** or **by voice**: an on-device speech-to-text program transcribes the child’s words. The transcript is shown to the child for review and then can be sent just like a regular text input.

3.2 Pip: a deliberately imperfect helper

Pip acts like a helper the child can interact with: the child’s description arrives through a conversation, a few details at a time. Pip is just a scribe: carry the child’s details in the child’s own words, typos and grammar errors included; add nothing the child did not say. Having a helper who can communicate makes it easy for the child to interact and disagree. The framing descends from virtual literacy peers, notably Sam, a virtual peer that alternated storytelling turns with children and whom children spontaneously addressed and coached [Ryokai, Vaucelle & Cassell, 2003]. The role here is less equal: Sam shares narration, while Pip just draws what it is told. Pip calls a chat model (gpt-5.4-mini) and an image model (gpt-image-2) through the proxy (§3.3).

Draw by default; ask at most one question: if the child’s initial message gives Pip enough detail, Pip generates the image immediately. If the request is too vague, such as "Draw a cat", Pip asks one clarifying question, such as "Fun! What color is the cat, or where is it?"

Complete-scene prompts; the picture carries the look: each picture is created and updated with all past instructions, to keep the characters and style consistent between images. The composed prompt

146 is created by stitching together every message the child has sent so far into one description, so the
147 image generator processes it as one request. The platform appends the same one-line constraint to
148 every drawing request. An example, with the content safeguard in italics:

149 Child: a sleepy koala librarian stamping tiny books at a desk
150 [Pip generates a picture]...
151 Child: add a lamp with a green shade on the desk
152 Drawing request sent to model after the second turn: a sleepy koala librarian stamping tiny books at a desk,
153 add a lamp with a green shade on the desk *This is an illustration for a children's storybook. Do not add*
154 *anything frightening, violent, gory, or adult that the description did not explicitly ask for — match the*
155 *intensity of the description and never exceed it. Absolutely no nudity or sexual content; no blood, wounds,*
156 *gore, injury or death; no real firearms or military hardware; nothing hateful or cruel. Keep it at the level of*
157 *a classic children's storybook for seven-to-nine-year-olds: mild adventure is fine; storybook props such as*
158 *a sword, shield or bow may be worn or carried but never aimed at, striking, or hurting anyone; no realistic*
159 *or horrifying creatures; nobody hurt or afraid.*

160 3.3 The proxy, safety, and custody

161 A small proxy holds the API key, relays the requests, and rate-limits requests from each client.
162 Personal information is automatically removed and blocked before anything gets sent. Character
163 names are retained because they are needed for story coherency. The server does not retain prompts
164 or images; it only relays them. Safety is layered, and different layers catch different failures. Pip's
165 contract tells it to refuse prohibited requests involving nudity, gore, or real firearms and to offer a
166 safer alternative. For example, when asked for "a kid holding a real pistol," Pip responds, "Hmm, no
167 weapons in my pictures — how about the kid holding a balloon or a lantern instead?" As a backstop,
168 a word filter on the drawing request path catches firearm and explosive terms. Every drawing request
169 also carries a one-line content safeguard, shown in place (§3.2). Section 4.2 evaluates whether the
170 content safeguards block prohibited material without suppressing ordinary story content for children
171 ages 7-9.

172 The project is saved as a versioned JSON file. It contains the story, the pages, the prompts, the images,
173 and the record of every change. During a session the project autosaves on the child's device. The
174 adult picks the save location, and the browser keeps a fallback copy. In class mode the encrypted file
175 passes through a relay to the supervising adult's device. The supervising adult can erase it afterward.
176 There are no child accounts and no server copy the platform can read.

177 Prompts pass through the proxy to the model provider, and images come back under its API terms.
178 Rate limiting uses the client's IP address. The proxy keeps no lasting IP log.

179 4 System evaluations

180 This section evaluates two parts of Story Sprout's image request pipeline: description fidelity in
181 the composed prompt and the content safeguard. Both use scripted inputs and the deployed models
182 without involving children. Full methods, scripts, data, and item-level results are available in the
183 GitHub repository (§1).

184 4.1 Description fidelity in the composed prompt

185 Pip is instructed to carry the child's wording and details into the composed image prompt without
186 correcting, adding, or omitting content. We tested this behavior with scripted conversations written
187 by the researchers, using the same models and APIs as the deployed studio. Each sequence included
188 four image requests, planted misspellings, and specific scene details. A 200-turn development suite
189 was used to refine the contract and select the chat model. The first model corrected the planted
190 misspellings, so it was replaced with gpt-5.4-mini. We then froze the contract, recorded its hash, and
191 tested it on 60 new sequences that had not appeared during development. Three runs produced 906
192 exchanges and 720 composed prompts. Every planted misspelling was preserved, and no prompt
193 added content that had not been supplied. Of the 720 prompts, 713 preserved all supplied content, or
194 99.0%. The remaining seven omitted at least one detail. All seven failures came from one four-turn
195 sequence in which a revision from "big" to "small and cozy" sometimes also removed details such as
196 "oak," "rope," and "swing." Pip followed the interaction rules in every exchange.

197 4.2 The content safeguard

198 Setting the content safeguard requires balancing three concerns. Children need room to author stories
199 that engage fear, conflict, sadness, and the forces of good and evil [Sutton-Smith, 1981]. Frightening
200 media encountered in childhood can also produce lasting reactions, with triggers and coping varying
201 by developmental stage [Harrison & Cantor, 1999]. At the same time, a system designed for children
202 has a responsibility to exclude clearly age-inappropriate material. We therefore implemented an
203 evaluator for the deployed image route, including the platform's word filter and appended content
204 safeguard (§3.2), with 80 scripted descriptions: 40 to keep, 25 to block, and 15 boundary cases. Each
205 ran three times with the safeguard and three times with only the provider's moderation [OpenAI,
206 2026]. The requested element was preserved in 110 of 120 keep set trials (91.7%); nine were blocked
207 by the word filter, all from three mild descriptions containing "blood," "bloody," or "bleeding," while
208 one was refused by the provider. In the block set, 59 of 75 trials ended without the prohibited element
209 (78.7%), while 16 trials from six descriptions still depicted it. In the boundary set, 36 of 45 trials
210 produced images; all nine refusals came from three firearm-word descriptions blocked by the word
211 filter. The word filter and provider moderation caught different failures, but neither was sufficient.
212 The trade-off between safety and preserving the creative and educational value of the activity warrants
213 further study.

214 5 Deployment and practice

215 The curriculum was initially inspired by the Cross-Media Projects at Harvard's X-Media Lab, which
216 use stories presented across multiple media to support interdisciplinary K–12 learning [X-Media
217 Lab, n.d.]. Two eight-session summer programs ran in 2025 and 2026, 75 minutes per session. A
218 ten-session after-school program ran in spring 2026, 30 minutes per session. The authors built the
219 platform across the 2025–26 school year. It entered pilot use in spring 2026 and was used in the 2026
220 summer program. The curriculum has reached over 100 children. Roughly 50 have used the deployed
221 platform. No child data was collected for this research. The observations here are practice notes, not
222 measurements.

223 Two curriculum decisions emerged from those sessions and correspondence with colleagues in the
224 field of child–computer interaction. The youngest children tended to accept the first illustration as-is.
225 Older children revised on their own. That observation set a provisional target of ages 7–9, rather than
226 the full elementary school range. The second decision changed the session framing. Children have
227 always read their finished books to peers. The read-aloud is now announced at the start as the point of
228 the task, making the picture book a public artifact. Rosebud, a storytelling system designed for ages
229 7–9, elicited a great deal of highly personal writing, but children showed little interest in rewriting
230 because the system gave them no intrinsic reason to revisit their text [Glos & Cassell, 1997; Cassell,
231 2004]. Story Sprout adds two reasons to continue: a mismatched picture invites further specification,
232 and an announced class read-aloud gives children a reason to finish the book.

233 6 Limitations, future work, and ethics

234 One limitation may persist as image models improve. Even a precise description may not produce the
235 image a child intends. A child who continues revising may eventually accept or adapt to the model's
236 interpretation, even when it does not match the scene originally imagined. Story Sprout responds by
237 retaining every image attempt in the Image Bank.

238 A planned study, following ethics review and appropriate consent procedures, will test and evaluate
239 the system's ability to achieve the goals outlined above. To that end, a companion manuscript in
240 preparation develops the learning framework.

241 Story Sprout is developed and used in nonprofit educational settings. Because it serves children under
242 13, its privacy design was informed by the COPPA law (16 C.F.R. pt. 312, as amended 2025). This is
243 not a claim of exemption from or compliance with that law. The paper reports aggregate participation
244 counts and practice observations only. It includes no identifiable child information, child-authored
245 work, or individual-level session data.

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Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 4 states the suite construction, run counts, and frozen contract hash for the scribe evaluation, and the description sets (keep, block, boundary) and run design for the safeguard evaluation. Full configurations (models, request parameters, item sets, scoring rules, scorer errata) are recorded in the released evaluation materials (Section 1).

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Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: All experiments are evaluations against the hosted model APIs named in Section 3.2; no model training or fine-tuning was performed, and no GPU or other specialized hardware is required. The scribe-contract evaluation (Section 4.1) is a 200-turn development suite plus three held-out runs over 60 new sequences, totaling 906 chat exchanges and 720 composed prompts. The content-safeguard evaluation (Section 4.2) is 480 image-route trials: 80 scripted descriptions, each run three times with the safeguard and three times without it. Running the evaluations requires no specialized compute beyond API access to those models.

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